

Chocolate Sales Data Warehouse

Project Documentation

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Role: Data Engineer / Data Analyst

Project Overview

This project demonstrates the implementation of an end-to-end Data Warehouse for Chocolate Sales data (2023–2024). The solution integrates PostgreSQL, Pentaho Data Integration (PDI), SQL, and Tableau to transform raw transactional data into analytical insights.

Objectives

- Build an ETL pipeline
- Design a Star Schema Data Warehouse
- Create a Data Mart
- Develop an interactive Tableau dashboard

Technology Stack

- PostgreSQL
- Pentaho Data Integration (PDI)
- DBeaver
- SQL
- Tableau Public
- Git & GitHub

Architecture

CSV Dataset -> PostgreSQL (OLTP) -> Staging -> ETL (Pentaho) -> Data Warehouse -> Data Mart -> Tableau Dashboard

Data Warehouse Design

Star Schema consisting of:

- Fact: fact_orders
- Dimensions: dim_customers, dim_product, dim_store, dim_date

Implemented SCD Type 2 for Customer and SCD Type 1 for Product, Store, and Date.

ETL Process

1. Extract data from OLTP.
2. Load to staging.
3. Clean and transform data.
4. Generate surrogate keys.

5. Load dimensions and fact table.
6. Build Data Mart.

Dashboard Insights

- Revenue: 25,486,129
- Profit Margin: 40%
- Quantity Sold: 2,999,589
- Dark Chocolate was the best-selling category.
- Ferrero and Cadbury were top-performing brands.
- Airport stores generated the highest sales.

My Contribution

- Designed the Data Warehouse.
- Developed SQL scripts.
- Built ETL workflows in Pentaho.
- Created the Data Mart.
- Built Tableau dashboard.
- Documented the project.

Repository Structure

README.md
docs/
images/
sql/
pentaho/
tableau/

Next Steps

Add SQL scripts, Pentaho jobs (.kjb/.ktr), Tableau workbook, dashboard screenshots, and architecture images into the repository.